CHAPTER TWENTY-ONE

# Apathy in Parkinson's Disease

**Javier Pagonabarraga**[*,†,‡,§,1], **Jaime Kulisevsky**[*,†,‡,§]
[*]Movement Disorders Unit, Sant Pau Hospital and Biomedical Research Institute (IIB-Sant Pau), Barcelona, Spain
[†]Universitat Autònoma de Barcelona, Barcelona, Spain
[‡]Centro de Investigación en Red-Enfermedades Neurodegenerativas (CIBERNED), Barcelona, Spain
[§]Universitat Oberta de Catalunya (U.O.C.), Barcelona, Spain
[1]Corresponding author: e-mail address: JPagonabarraga@santpau.cat

## Contents

1. The Concept of Apathy 657
2. The Clinical Relevance and Scientific Interest of Defining Apathetic Syndromes 659
3. Apathy in PD: Frequency and Distribution of Apathetic Syndromes 662
4. Diagnosis of Apathy in Clinical Practice: Scales and Diagnostic Criteria for PD 667
5. Treatment of Apathy in PD 669
6. Conclusions 670
References 672

## Abstract

The normal maintenance of human motivation depends on the integrity of subcortical structures that link the medial and lateral prefrontal cortex with the limbic system.

Apathy is highly prevalent in Parkinson's disease and causes major impact on the quality of life of patients and caregivers, comparable to depression or cognitive impairment. The clinical differentiation of apathy from the emotional symptoms of depression, and from difficulties in planning or organizing mental programs as a consequence of executive dysfunction, may guide a rationale for individualized treatment approach of apathetic symptoms, which is presently lacking. To review the different apathetic syndromes that can be diagnosed in clinical practice by appropriate scales, as well as the brain systems that subserve each syndrome, helps to explain how dopaminergic, antidepressant, or cholinergic medications may lead to individual improvements in apathy.

## 1. THE CONCEPT OF APATHY

Apathy is a behavioral syndrome that refers to a set of simultaneous behavioral, affective, and cognitive features variably characterized by reduced interests, emotions, and motivations, not attributable to diminished level of consciousness, cognitive impairment, or emotional distress

*International Review of Neurobiology*, Volume 133
ISSN 0074-7742
http://dx.doi.org/10.1016/bs.irn.2017.05.025
© 2017 Elsevier Inc.
All rights reserved.

(Marin, 1991). Apathy may be driven by decreased cognitive interests, emotional disturbances such as depression or flattening affect, or by deficits in spontaneous activation of mental processes (autoactivation deficit) (Starkstein, 2012).

In the last years, an effort has been made to describe mood disorders in Parkinson's disease (PD) in more detail, trying to separate better the presence and severity of apathy, depression, and anhedonia (Kirsch-Darrow, Marsiske, Okun, Bauer, & Bowers, 2011; Zahodne, Marsiske, Okun, & Bowers, 2012). Isolated apathy not related to cognitive impairment or depressive symptoms has been observed in both early and advanced PD patients (Funkiewiez, Ardouin, Caputo, et al., 2004; Martinez-Horta, Pagonabarraga, Fernandez de Bobadilla, Garcia-Sanchez, & Kulisevsky, 2013). Beyond nosological explanations that help to differentiate different neural substrates related to each one of these types of apathy, a more detailed classification of apathetic syndromes entails therapeutic advantages, helping to differentiate patients more likely to improve by using dopaminomimetics, antidepressants, or cholinesterase inhibitors (Brown & Pluck, 2000). Differentiation from depression, and discrimination of its cognitive, emotional, and autoactivation components may guide a rationale for individualized treatment approach of apathetic symptoms.

After reviewing initial descriptions, the conceptual history, and prevalence and clinical relevance of apathy in PD, we propose a model that links apathetic syndromes with different etiologies and neurobiological correlates, thus providing an operative framework for classifying patients better and optimizing treatment.

Motivation is a psychological feature that arouses an organism to act toward a desired goal, both eliciting and sustaining goal-directed behaviors (Kuhl & Atkinson, 1986). Motivation in healthy human beings acts as an inner drive elicited by physical needs (i.e., eating, drinking), but also by pleasurable and aversive stimuli, as coded by reward- and punishment-based learning, that "stamps-in" motivational importance to otherwise neutral environmental stimuli (McCutcheon, Ebner, Loriaux, & Roitman, 2012; Mellers, Schwartz, & Cooke, 1998; Mogenson, Jones, & Yim, 1980).

The concept of apathy as a clinical syndrome was built from original case reports and lesional studies describing patients with relatively well-preserved cognition showing a striking poverty of observable active behaviors, but who were still capable to react and behave almost normally only if and when stimulated. This unique combination of lack of self-initiated goal-directed actions, giving the patient the aspect of indifference, flattening of affect,

or emptiness of mind, and diminished but still obtained responsiveness to stimuli, has been named with many descriptive epithets. Naville (1922) introduced the term "bradyphrenia" to describe the lack of interest, initiative, and concentration observed in patients following pandemic encephalitis lethargica. Patients were observed "to sit and do nothing unless exhorted to rouse themselves."

Cairns, Olfield, Pennybacker, and Whitteridge (1941) termed "akinetic or trance-like mutism" to "a peculiar mental state … due to pressure of a cyst upon structures surrounding the 3rd ventricle" that caused "absence of a *normal stream of impulses* to the cortex … leading the patient to lie inert, without emotional expression," and responding in a "feeble, slow, and incomplete manner" after "repeated commands" only.

Both, akinetic mutism and "abulia," used for severe deficits in spontaneous activation of mental processes, have been described after lesions restricted to the anterior cingulate cortex (ACC), caudate nucleus, supplementary motor area, and midbrain/diencephalic lesions (Fisher, 1983; Nagaratnam, Nagaratnam, Ng, & Diu, 2004; Vijayaraghavan, Krishnamoorthy, Brown, & Trimble, 2002). "Autoactivation deficit," "pure psychic akinesia," or "athymhormia" have been used to stress the reversal of psychic autoactivation deficit by external stimulation (Habib, 2004; Laplane, Baulac, Widlocher, & Dubois, 1984; Laplane & Dubois, 2001). Patients with psychic akinesia report "a striking absence of thoughts or spontaneous mental activity," "stay motionless and speechless during endless periods," and "express the feeling that their mind is empty when they are not stimulated" (Laplane et al., 1984). After external stimulation, patients may be engaged in complex behaviors with almost normalized cognitive performance.

Applied to patients with discrete brain lesions presenting different states of diminished drive and motivation, all the previously referred terms can be considered as describing a continuum of severity of reduced behaviors, from the most severe forms of akinetic-mutism, to abulia, and athymhormia/autoactivation deficits. In neurodegenerative diseases, decreased goal-directed behaviors are generally categorized as "apathy," regardless of their severity.

## 2. THE CLINICAL RELEVANCE AND SCIENTIFIC INTEREST OF DEFINING APATHETIC SYNDROMES

The normal maintenance of human motivation depends on the integrity of subcortical structures that link the medial and lateral prefrontal cortex with the limbic system. Apathy is a clinically relevant and potentially

treatable dimension of behavior (Marin, Fogel, Hawkins, Duffy, & Krupp, 1995). Conceptually, apathy can stand from the variable contribution (isolated, or more frequently in combination) of four major components of goal-directed behavior: reward deficiency syndrome, emotional distress, executive dysfunction, and autoactivation failure.

Dysfunction of circuits connecting—through the ACC—the orbitomedial and ventromedial PFC with the amygdala, ventral striatum, and hypothalamus (Groenewegen, Wright, & Uylings, 1997; Mega, Cummings, Salloway, & Malloy, 1997) leads to emotional blunting or lack of emotional resonance, which prevents the subject from "stamping-in" motivational values to environmental or inner stimuli. A reward deficiency syndrome would then be produced by degeneration of these structures as in frontotemporal dementia or progressive supranuclear palsy, or by mesolimbic and mesocortical dopamine denervation in patients with degeneration of the midbrain dopaminergic neurons in the substantia nigra (SN) and ventral tegmental area (VTA), as in PD (Drui, Carnicella, Carcenac, et al., 2014; Ilango et al., 2014).

Blunted affect should be differentiated from apathy caused by negative effect. In major depression, sadness is accompanied by a marked reduction in effortful and sustained positively motivated behaviors (Liao, Feng, Zhou, et al., 2012). In some patients, however, apathy may persist despite improvement in sadness, and in these cases residual blunted affect relates to decreased functionality in fronto-limbic brain circuits (Robinson, Cools, Carlisi, Sahakian, & Drevets, 2012). There is a large overlap between apathy and depression, with a series of common features. Depression can include apathy, but apathy can also present as an isolated syndrome. Several symptoms are specific to depression and their presence or absence helps to separate depression from isolated apathy (Table 1). Clinical diagnosis of apathy should then be based on the presence of symptoms causing decreased goal-directed behaviors, that can be present in isolated, or in combination with either depressive or cognitive/dysexecutive symptoms.

Numerous studies have shown that executive dysfunction is the most consistent neuropsychological correlate of apathy in neurodegeneration (Brown & Pluck, 2000; Jahanshahi, 1998). Executive dysfunction is related to decreased activity in a network that encompasses the dorsolateral PFC (DLPFC), lateral aspects of the caudate nucleus and putamen, ACC, and posterior parietal cortex (Eckert, Tang, & Eidelberg, 2007). The dysfunction of this network makes it difficult to redirect attention to novel stimuli, manipulate complex external or internal information, or generate plans

**Table 1** Clinical Differentiation Between Apathetic Symptoms and the Emotional Symptoms of Depression

| Apathetic Symptoms | Overlapping Symptoms | Emotional Symptoms of Depression |
|---|---|---|
| – Reduced initiative<br>– Loss of interest in social events or everyday activities<br>– Participative only if another person is engaging her/him<br>– Decreased interest for starting new activities<br>– Decreased interest in the world around him/her<br>– Emotional indifference<br>– Diminished emotional reactivity<br>– Less affectionate<br>– Lack of concern | – Psychomotor retardation<br>– Anhedonia<br>– Anergia<br>– Less active than usual<br>– Less enthusiastic about his/her usual interests | – Sadness<br>– Guilt feelings<br>– Negative thoughts and feelings<br>– Helplessness<br>– Hopelessness<br>– Pessimism<br>– Self-criticism<br>– Anxiety<br>– Suicidal ideation |

for the future, leading to a state of *cognitive inertia* that limits the search for, or the enjoyment of previous or new interests (Levy & Dubois, 2006).

Finally, apathy may also result from an inability to activate oneself. As mentioned earlier, bilateral lesions involving subcortical structures that link both the medial and lateral PFC with the limbic system (including the amygdala and ventral striatum) may lead to a state of autoactivation deficit. This connection is basically driven by the ACC (Habib, 2004), which plays a major role in regulating the expression of emotional responses, and to associate emotions with complex stimuli (Bush, Luu, & Posner, 2000; Etkin, Egner, & Kalisch, 2011). Disconnection of frontal regions with limbic subcortical nuclei (Mega & Cohenour, 1997) would prevent the subject to be engaged with both the external world and the *milieu intérieur* (Gross, 2009), leading to a state of "mental emptiness" in which the patient has "nothing to say" and "nothing matters."

These neural systems are richly innervated by the mesolimbic–mesocortical pathway and its dopaminergic afferents coming from the SNc and VTA, and by the nigrostriatal dopamine system that provides dopamine to the putamen, dorsolateral caudate, and PFC (Parent & Hazrati, 1995). However, serotoninergic, cholinergic, and noradrenergic neurotransmitter systems are also critical for the functionality of these circuits interconnecting the amygdala, ventral striatum, and PFC (Gray, 1999).

## 3. APATHY IN PD: FREQUENCY AND DISTRIBUTION OF APATHETIC SYNDROMES

Apathy is a frequent neuropsychiatric disturbance, that may be present from the early disease stages in up to 25% of PD patients (Aarsland, Bronnick, Alves, et al., 2009; Barone, Antonini, Colosimo, et al., 2009), increasing to 40% in nondemented patients after 5–10 years of the disease (Dujardin, Sockeel, Devos, et al., 2007; Kulisevsky, Pagonabarraga, Pascual-Sedano, Garcia-Sanchez, & Gironell, 2008; Pluck & Brown, 2002), and to 60% when patients become demented (Aarsland, Bronnick, Ehrt, et al., 2007). Apathy has been mainly described in the context of progressive cognitive deterioration, but has been also found to manifest as a separate and isolated behavioral symptom from the earliest stages (Kirsch-Darrow, Fernandez, Marsiske, Okun, & Bowers, 2006; Martinez-Horta et al., 2013). Multivariate analysis has found apathy to independently predict lower functioning in activities of daily living (Laatu, Karrasch, Martikainen, & Marttila, 2013), decreased response to treatment, poor outcome, and worse quality of life (Barone et al., 2009; Dujardin, Sockeel, Delliaux, Destee, & Defebvre, 2009; van Reekum, Stuss, & Ostrander, 2005), and has also a major impact on caregiver emotional distress (Schiehser, Liu, Lessig, et al., 2013).

Specific neurotransmitter deficits and pathological changes contribute on making PD a paradigmatic disease in which apathy is multidimensional (Zahodne, Gongvatana, Cohen, Ott, & Tremont, 2013; Zahodne, Marsiske, et al., 2012).

Both, dopaminergic denervation and cholinergic abnormalities, affect the lateral and medial regions of the prefrontal cortex. Degeneration of mesencephalic dopamine neurons at Braak stage III imposes dopamine depletion to the mesocortical and mesolimbic system (Braak et al., 2003), and progression of the synucleinopathy to dopamine projection areas further impair their functionality during Braak stages IV and V (Goetz, Emre, & Dubois, 2008; Luk, Kehm, Carroll, et al., 2012). Delineation of the different components that may lead to apathy is then crucial to plan therapeutic strategies. Numerous studies have shown a robust association between the degree of apathy (but not depression) and the degree of executive dysfunction in PD (Isella, Melzi, Grimaldi, et al., 2002; Pluck & Brown, 2002; Starkstein et al., 1992). Nevertheless, when PD patients are assessed using more comprehensive neuropsychological batteries, a wider spectrum of

deficits appears in apathetic compared to nonapathetic patients (Dujardin et al., 2009; Martinez-Horta et al., 2013). Besides having more executive dysfunction, apathetic compared to nonapathetic patients exhibit significant differences on global cognitive function and on the performance of cognitive tasks more dependent on the temporal lobes (e.g., delayed verbal memory, semantic verbal fluency, naming) (Dujardin et al., 2009; Martinez-Horta et al., 2013). This helps to explain why apathy seems to herald dementia in PD (Dujardin et al., 2009). Furthermore, in studies that included highly selected patients with isolated apathy without concomitant depression or dementia, apathy appeared also related (i) to facial emotion recognition deficits—indicating dysfunction in mesolimbic circuits (Martinez-Corral, Pagonabarraga, Llebaria, et al., 2010; Robert, Le Jeune, Dondaine, et al., 2014), and (ii) to a better performance on decision-making tasks. This latter apparently paradoxical "protective" effect was suggested to be related to a more extensive degeneration of the VTA in apathetic patients that attenuate the "overdosing" effect associated to excessive dopaminergic stimulation from antiparkinsonian drugs in frontal areas less affected by dopaminergic denervation in PD, such as the ventral striatum (Kulisevsky et al., 1996; Martinez-Horta et al., 2013). Neurophysiological studies measuring the functionality of the ventral striatum and medial PFC by the feedback-related negativity wave (Carlson, Foti, Mujica-Parodi, Harmon-Jones, & Hajcak, 2011) reinforce the role of emotional blunting in the development of isolated apathy in PD (Martinez-Horta, Riba, de Bobadilla, et al., 2014). The amplitude of the feedback-related negativity—an event-related brain potential associated with performance outcome valence—was significantly reduced in nondemented PD patients with isolated apathy compared with both nonapathetic patients and healthy controls, indicating that apathy in this subset of patients results from impaired incentive processing and a compromised mesocorticolimbic pathway (Martinez-Horta et al., 2014). More recently, to further investigate, the structural brain changes involving reward circuits that associate with nondepressive apathy in PD patients, both the presence and severity of isolated apathy were related to gray matter volume loss in the left nucleus accumbens, a subcortical structure that is a key node of the human reward circuit (Martinez-Horta, Sampedro, Pagonabarraga, et al., 2016) (Fig. 1).

The multidimensional nature of apathy in PD is also clearly evidenced in studies that have comprehensively assessed the presence of depression, apathy, anhedonia, and cognition in nondemented PD patients (Cubo, Benito-Leon, Coronell, & Armesto, 2012; Kirsch-Darrow et al., 2011; Zahodne,

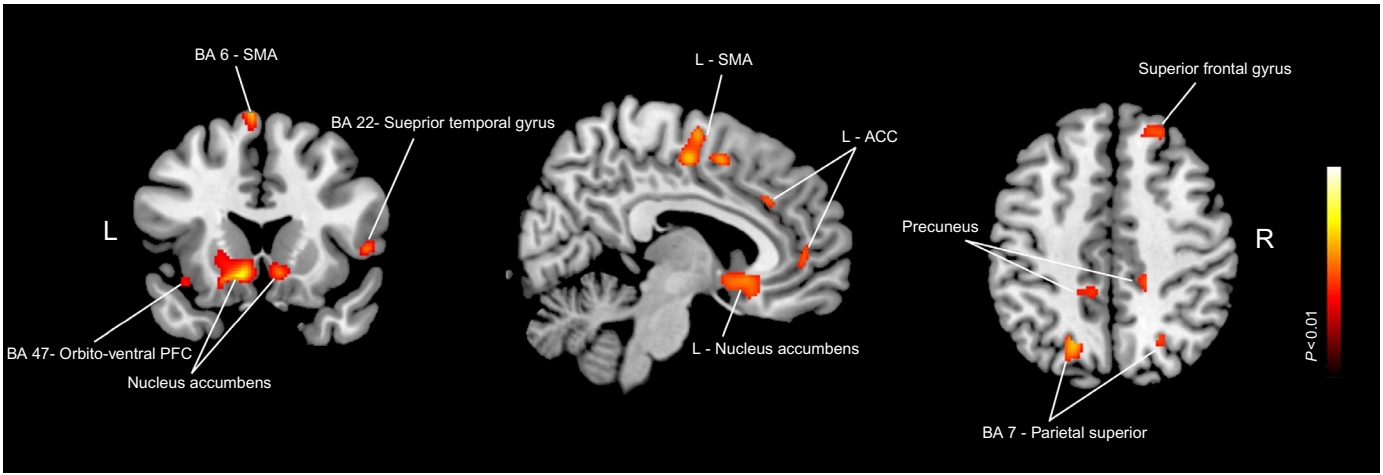


**Fig. 1** Apathy in nondepressed and nondemented PD patients correlated with gray matter volume loss in nucleus accumbens bilaterally (left > right), stressing the importance of mesolimbic degeneration in motivational deficits developed in PD.

Marsiske, et al., 2012; Zahodne & Tremont, 2013). By dissecting the different affective symptoms that may coexist in PD, anhedonia appears common both in apathy and depression, making it useless for differential diagnosis in clinical practice (Kirsch-Darrow et al., 2011; Zahodne, Marsiske, et al., 2012). Conversely, apathy seems specifically related to blunted affect, decreased intellectual curiosity, and executive dysfunction, while depression appears linked to the presence of sadness, dysphoria, negative emotions, and increased anxiety (Kirsch-Darrow et al., 2011; Zahodne, Marsiske, et al., 2012).

The use of specific rating instruments for each one of these domains (Leentjens, Dujardin, Marsh, et al., 2008a, 2008b; Schrag, Barone, Brown, et al., 2007) may help to disentangle in the single patient the different aspects contributing to decreased goal-directed behaviors.

A dysfunction of dopaminergic transmission in the mesocorticolimbic pathway has been significantly associated with apathetic behaviors in PD. Positron emission tomography (PET) studies with D2/D3 receptor antagonist, [$^{11}$C]-raclopride, have shown several differences in dopaminergic binding and transmission between apathetic and nonapathetic PD patients (Thobois, Ardouin, Lhommee, et al., 2010). In particular, [$^{11}$C]-raclopride binding potential (BP) was increased in apathetic PD patients in the OFC, cingulate cortex, DLPFC, amygdala, as well as in the striatum, implying either (reactive) increase in D2/D3 receptor expression and/or reduction in endogenous synaptic dopamine. Furthermore, the methylphenidate challenge led to a smaller reduction in [$^{11}$C]-raclopride BP in apathetic PD providing evidence for weaker capacity of endogenous dopamine release in the mesocorticolimbic system of these patients. Additional PET studies in PD patients with [$^{11}$C]RTI-32, a ligand with affinity to both dopamine and noradrenaline transporters, demonstrated that the degree of apathy severity was inversely correlated with [$^{11}$C]RTI-32 binding in the ventral striatum (Remy, Doder, Lees, Turjanski, & Brooks, 2005). Overall, these neuroimaging studies suggested that apathy in PD may result from a severe hypodopaminergic syndrome in the mesocorticolimbic system, leading to an impaired emotion reactivity and poor decision-making processes (Bechara, Damasio, & Damasio, 2000).

In alternative to the "emotional-affective" syndrome, apathy may also be associated with a disruption of "cognitive functions." This form is generally associated with an impairment in those executive functions (e.g., selection, planning, working memory, set-shifting) needed to elaborate an action plan. The "cognitive" apathy may result from brain lesions affecting either the

lateral PFC (i.e., dorsolateral and ventrolateral) or the caudate nucleus (Levy & Dubois, 2006). Several functional imaging studies in nondemented PD patients have supported this hypothesis showing that these patients may present with abnormal activations at the level of the dorsolateral/ventrolateral PFC and caudate nucleus during executive tasks (Monchi et al., 2004; Monchi, Petrides, Mejia-Constain, & Strafella, 2007). Glucose metabolism studies with (Laplane et al., 1984) FDG-PET have shown in PD that apathy severity correlated with metabolic activity in different cognitive regions including the inferior–middle frontal gyrus, cingulate cortex, insula, cuneus, as well as temporoparietal region (Huang, Ravdin, Nirenberg, et al., 2013; Le Jeune, Drapier, Bourguignon, et al., 2009; Robert, Clairet, Benoit, et al., 2002).

In the severe form of apathy previously referred as "autoactivation" deficit, the combined basal ganglia, thalamic, and cortical lesions influencing the dorsal–medial PFC (i.e., supplementary motor area, ACC) generally affect both cognitive and emotional neural networks (Levy & Dubois, 2006). When apathy coexists with major depression and sadness, it is accompanied by a marked reduction in effortful and sustained positively motivated behaviors that correlate with a dysfunctional network showing subgenual cingulate cortex (BA25) hyperactivity, along with hypometabolism of both the DLPFC and dorsal ACC (BA24b) (Goodwin, 1997; Mayberg, Liotti, Brannan, et al., 1999; Stone, Lin, & Quartermain, 2008). With recovery from depression, metabolism in dorsal ACC is restored and improvements in goal-directed behaviors observed (Buchsbaum, Wu, Siegel, et al., 1997; Mayberg, Brannan, Tekell, et al., 2000).

The combined effect of dopaminergic stimulation defects and degenerative changes within the limbic system is clearly exemplified by the analysis of apathy developed after deep brain stimulation of the subthalamic nucleus (STN-DBS). Postoperative apathy occurring after STN-DBS during the first postoperative year in relatively young L-dopa-sensitive PD patients, that are characterized by a predominant degeneration of midbrain dopaminergic neurons in the SN and VTA but without significant cortical synucleinopathy, can be seen as a model of relatively pure dopamine-dependent apathy. In this group of patients, modulation of the mesocorticolimbic networks by dopaminergic drugs (especially D2–D3 dopaminergic agonists with relatively high affinity to mesolimbic system) (Ye, Hammer, Camara, & Munte, 2011) improves the appetitive drive to perform pleasurable activities (Kulisevsky, Pagonabarraga, & Martinez-Corral, 2009; Lhommée et al., 2014). Conversely, decrease of dopaminergic

agonists after surgery, or the switch from dopamine agonists to levodopa, has been associated with the emergence of severe apathy, which can be reversed after a careful titration of the dose of the dopamine agonist (Kulisevsky et al., 2009).

## 4. DIAGNOSIS OF APATHY IN CLINICAL PRACTICE: SCALES AND DIAGNOSTIC CRITERIA FOR PD

Frequencies of apathy in PD are largely variable between studies, with reported frequencies ranging from 15% to 70% (Kirsch-Darrow et al., 2006; Kulisevsky et al., 2008; Pluck & Brown, 2002; Starkstein et al., 1992). These differences seem mainly based on the diagnostic approach used, whether the diagnosis of apathy has been established according to cut-off scores on severity rating scales, instruments rated by caregivers, or clinical diagnostic criteria.

The multidimensional nature of apathy and the subjective nature of this neuropsychiatric symptom make it difficult to be accurate in its diagnosis and to develop tools or criteria operative enough to be generalizable between physicians and research groups.

The development of specific scales for apathy has been one of the most important advances in the recognition of this problem both in the clinical practice and in the research field. A commissioned task force by the Movement Disorders Society reviewed the clinimetric properties of apathy in PD samples (Leentjens et al., 2008a). Based on the use of the scale in research studies, use beyond original developers, successful clinimetric properties, and the presence of cut-off scores for screening of apathy in PD, only the Apathy Scale (AS) (Starkstein et al., 1992) was classified as "recommended." Although the UPDRS item 4 also met the criteria to be classified as recommended, it should be considered for screening only. Scales that met criteria to be classified as suggested were the Apathy Evaluation Scale (AES) (Marin, Biedrzycki, & Firinciogullari, 1991), the Lille Apathy Rating Scale (LARS) (Sockeel et al., 2006), and the item for apathy (item 7) in the Neuropsychiatric Inventory (NPI) (Cummings et al., 1994). The LARS has been used in PD beyond original developers (Zahodne, Young, Kirsch-Darrow, et al., 2009), so this scale would currently meet criteria for recommended. The LARS is an interesting scale that, based on Marin's original conceptualization, provides an overall apathy score, and four composite subscores that assess relevant subdimensions of apathy: intellectual curiosity, action initiation, emotion, and self-awareness.

However, the task force considered that the lack of widely accepted diagnostic criteria for apathy (i) hampers the development of valid assessment scales because no gold standard is available and (ii) is a major barrier to facilitate the study of this nosological entity across different neuropsychiatric disorders and in relation to depression and dementia (Leentjens et al., 2008a). And especially now that evidence is emerging that some forms of pharmacotherapy may be specifically beneficial for motivational symptoms, widely accepted criteria would be necessary to register new medications for this indication (Leentjens et al., 2008a).

Regarding diagnostic criteria, only two studies have published formal validations of clinical criteria specific for PD (Starkstein et al., 2009) or criteria applicable for AD and other neuropsychiatric disorders (Drijgers, Dujardin, Reijnders, Defebvre, & Leentjens, 2010; Robert, Onyike, Leentjens, et al., 2009). Specific clinical criteria for apathy in PD proposed by Starkstein et al. are based on diminished motivation relative to the patient's previous level, and require the presence of at least one symptom belonging to each of the following domains: diminished goal-directed behavior, diminished goal-directed cognition, or diminished emotional response to positive or negative events (Starkstein et al., 2009). In a sample of 164 PD patients, 32% met criteria for apathy. Given the demographic and clinical features of this sample, with a mean age of 68 years, mean MMSE = 24.2, and 53% with Hoehn & Yahr $\geq$ III, this frequency seems lower than expected, which could be explained by the criteria's requirement to present at least one symptom belonging to each domain.

Robert et al. proposed diagnostic criteria for apathy to be applied in AD and other neuropsychiatric disorders (Robert et al., 2009). The authors also propose three symptomatic domains that cover behavioral, cognitive, and emotional aspects of apathy, but are less restrictive for the final diagnosis of apathy, requiring the presence of at least one symptom in two of these three domains. The application of Robert's criteria in a sample of 122 PD patients also disclosed a low frequency of apathy. In a sample with mean age of 64 years and 8.5 years of disease duration, the frequency of apathy was 17.2%, also lower than expected as compared with the frequency of apathetic symptoms in similar samples that used less restrictive scales such as the NPI and the NMSS (Barone et al., 2009; Kulisevsky et al., 2008; Martinez-Martin, Rodriguez-Blazquez, Kurtis, & Chaudhuri, 2011).

In fact, these diagnostic criteria have not been widely used in recent years, which could be due to the abstract nature of the items included in each domain, and the difficulty of its application in daily clinical practice. In

addition, although criteria are covering the most important subdimensions of apathy, no items are included that may allow to relate apathetic symptoms to their possible etiology: depression/emotional distress, cognitive impairment, or reward deficiency syndrome.

## 5. TREATMENT OF APATHY IN PD

Pharmacologic interventions with cholinesterase inhibitors, dopaminergic drugs (including methylphenidate), and antidepressants have been tried to improve apathy associated with PD.

To test the efficacy of pharmacological interventions for apathy in PD, two randomized controlled trials including apathy as the primary outcome variable have been conducted. In a double-blind, placebo-controlled study of 31 PD patients with moderate to severe apathy without dementia and depression, apathy was assessed using the LARS and significantly improved after 6 months on treatment with rivastigmine 9.5 mg/day ($P=0.031$; adjusted size effect: −0.9) (Devos, Moreau, Maltete, et al., 2014) (Level B, 1 Class I).

To explore the usefulness of dopamine agonists to treat apathy in PD, the D2/D3 dopamine agonist piribedil significantly improved apathy in 37 patients with PD presenting with apathy following STN-DBS, as assessed by the Starkstein AS score (AS > 14). In this 12-week prospective, placebo-controlled, randomized, double-blinded trial, the SAS score reduced significantly by 34.6% in the piribedil group ($n=19$), compared with 3.2% on placebo ($n=18$) ($P=0.015$) (Thobois, Lhommee, Klinger, et al., 2013) (Level B, 1 Class I). In an open-label study, the D2/D3 dopamine agonist ropinirole, proved to be efficacious in eight patients with apathy following STN-DBS, as manifested by significant decreases in AS scores after 6 months of treatment (Czernecki, Schupbach, Yaici, et al., 2008). Also in open-label studies, and using nonspecific scales for apathy, pramipexole and rotigotine have shown significant improvements in the motivational items of the UPDRS-I and the Nonmotor Symptoms Scale (Leentjens et al., 2009; Ray Chaudhuri, Martinez-Martin, Antonini, et al., 2013) (Level C, 1 Class II and 2 Class III studies).

The potential benefit of the dopaminergic drug methylphenidate on parkinsonian apathy has been reported in PD in just a single case (Chatterjee & Fahn, 2002) (Level U) but is in line with recent double-blind, placebo-controlled studies in AD that have shown mild but significant improvement of apathy methylphenidate 20 mg/day (Herrmann, Rothenburg, Black,

et al., 2008; Lanctot, Chau, Herrmann, et al., 2014). Methylphenidate is known to enhance mesolimbic dopaminergic stimulation by relatively selective inhibition of the dopamine transporter and has been abused as a recreational drug to enhance emotional motivational aspects of human behavior. In fact, in the prelevodopa era, intravenous and oral administration of methylphenidate produced a lightening of mood and euphoria in PD (Halliday & Nathan, 1961). In this randomized crossover trial of oral methylphenidate, half of the patients felt better on methylphenidate. These beneficial effects were not derived from improvement in antiparkinsonian features, but from a subjective increase in energy and lightening of their mood (Halliday & Nathan, 1961). Of note, parkinsonian patients with major depression exhibited a significant lack of sensitivity to the euphoriant effects of methylphenidate, in comparison with PD without major depression or healthy controls (Cantello, Aguggia, Gilli, et al., 1989).

The use of antidepressants for apathy is controversial. Single cases of postoperative parkinsonian apathy resistant to antidepressants have been reported. And notably, SSRIs have been related to increased apathy in PD (Zahodne, Bernal-Pacheco, Bowers, et al., 2012) and among depressed elderly (Wongpakaran, van Reekum, Wongpakaran, & Clarke, 2007), while the norepinephrine–dopamine reuptake inhibitor bupropion has been reported to improve motivation (Corcoran, Wong, & O'Keane, 2004).

## 6. CONCLUSIONS

The normal maintenance of spontaneous mental activity, the capacity to associate emotions with complex stimuli, and the adequate expression of these emotions, depends on the integrity of subcortical structures that link the medial and lateral prefrontal cortex with the limbic system.

Discrete lesions of these structures, including the amygdala and the ventral striatum and its main connection through the ACC, have been associated with the acute appearance of severe deficits in spontaneous stimulation of mental processes and psychic autoactivation. These descriptions have led to the current clinical concept of apathy, as a behavioral disorder with reduced motivation translating as decrease in goal-oriented behavior with both cognitive and emotional aspects (Marin, 1991).

The observer is confronted with individuals with reduced interests, emotions, and motivations that still have the capacity to describe their poverty of spontaneous thoughts and feelings which can partially improve during external stimulation.

Neurodegenerative lesions of the ventromedial PFC, amygdala, and ventral striatum produce similar disruption of emotional-affective and linked cognitive mechanisms (mainly dysexecutive syndrome) that adjusts to the clinical construct of apathy (Marin, 1996). In AD, PD, PSP, HD, and FTD, apathy can be clearly separated from other symptoms and measured with adequate instruments.

Apathy can be subclinical and only evidenced during purposeful examinations (mostly in AD and PD), or can be prominent from the beginning (mostly in PSP, HD, and FTD). Nevertheless, severity of apathy frequently aggravates along the course of all these diseases in which constitutes an independent predictor of poorer outcome (Dujardin et al., 2009; Hurt, Bhattacharyya, Burns, et al., 2008; van Reekum et al., 2005) and has a major impact on caregiver distress (Merrilees et al., 2013). Neuroimaging studies in neurodegenerative diseases have linked the state of "cognitive inertia" characteristic of apathy (Levy & Dubois, 2006) to decreased activity in a network that encompasses the DLPF cortex, lateral aspects of the caudate nucleus and putamen, ACC, and posterior parietal cortex (Benoit et al., 2002; Eckert et al., 2007; Lanctot, Moosa, Herrmann, et al., 2007; Marshall et al., 2007; Migneco, Benoit, Koulibaly, et al., 2001). Symptoms of apathy are frequent in major depression and may even persist after improvement in sadness with observed decreased functionality in similar fronto-limbic brain circuits (Liao et al., 2012) that is restored when improvements in goal-directed behaviors are observed (Buchsbaum et al., 1997; Mayberg et al., 2000).

We have addressed PD, as a paradigmatic situation in which the symptoms of apathy associated to deficit on dopaminergic innervation from mesolimbic and mesocortical can be manipulated to the extremes of its clinical spectrum by drugs and functional procedures commonly used to ameliorate the motor symptoms of the disease. While PD patients exposed to non-physiological dopaminergic stimulation are at high risk of developing a range of behavioral addictions or ICDs (Ray, Miyasaki, Zurowski, et al., 2012; Steeves, Miyasaki, Zurowski, et al., 2009; van Eimeren, Pellecchia, Cilia, et al., 2010), they are also exposed to the acute symptoms of apathy associated to the dopaminergic withdrawal syndrome (Castrioto, Lhommee, Moro, & Krack, 2014; Lhommee, Klinger, Thobois, et al., 2012; Pondal, Marras, Miyasaki, et al., 2013; Thobois et al., 2010) or to the more delayed appearance of apathy associated to the reduction of dopaminergic treatments after DBS (Drapier, Drapier, Sauleau, et al., 2006; Funkiewiez et al., 2004; Krack, Pollak, Limousin, et al., 1998; Lhommee et al., 2012).

Besides amelioration of the apathetic symptoms associated to the withdrawal of dopaminergic drugs with the reintroduction of dopaminergic stimulation (Thobois et al., 2013), no consistent pharmacological response of apathy has been observed in many exploratory studies. Cholinesterase inhibitors, dopaminergic drugs, methylphenidate, and antidepressants have all been tried in neurodegenerative disorders in studies that were either open-label or in which apathy was not the main outcome variable and was not explored with adequate instruments (Boyle & Malloy, 2004; Drijgers, Aalten, Winogrodzka, Verhey, & Leentjens, 2009). Recent double-blind, placebo-controlled studies showed promising results using rivastigmine for apathetic PD patients without dementia and depression (Devos et al., 2014), and methylphenidate for improving apathy in AD (Lanctot et al., 2014; Padala, Burke, Shostrom, et al., 2010). Altogether, there is good rationale for both dopaminergic (L-dopa, dopamine agonists, amphetamines, and dopamine-reuptake inhibitors) to improve mainly emotional, but also cognitive aspects of motivation, and for cholinergic (cholinesterase inhibitors) enhancement in the treatment of cognitive aspects of apathy.

## REFERENCES

Aarsland, D., Bronnick, K., Alves, G., et al. (2009). The spectrum of neuropsychiatric symptoms in patients with early untreated Parkinson's disease. *Journal of Neurology, Neurosurgery, and Psychiatry*, *80*, 928–930.

Aarsland, D., Bronnick, K., Ehrt, U., et al. (2007). Neuropsychiatric symptoms in patients with Parkinson's disease and dementia: Frequency, profile and associated care giver stress. *Journal of Neurology, Neurosurgery, and Psychiatry*, *78*, 36–42.

Barone, P., Antonini, A., Colosimo, C., et al. (2009). The PRIAMO study: A multicenter assessment of nonmotor symptoms and their impact on quality of life in Parkinson's disease. *Movement Disorders*, *24*, 1641–1649.

Bechara, A., Damasio, H., & Damasio, A. R. (2000). Emotion, decision making and the orbitofrontal cortex. *Cerebral Cortex*, *10*, 295–307.

Benoit, M., Koulibaly, P. M., Migneco, O., Darcourt, J., Pringuey, D. J., & Robert, P. H. (2002). Brain perfusion in Alzheimer's disease with and without apathy: A SPECT study with statistical parametric mapping analysis. *Psychiatry Research*, *114*, 103–111.

Boyle, P. A., & Malloy, P. F. (2004). Treating apathy in Alzheimer's disease. *Dementia and Geriatric Cognitive Disorders*, *17*, 91–99.

Braak, H., Del Tredici, K., Rub, U., de Vos, R. A., Jansen Steur, E. N., & Braak, E. (2003). Staging of brain pathology related to sporadic Parkinson's disease. *Neurobiology of Aging*, *24*, 197–211.

Brown, R. G., & Pluck, G. (2000). Negative symptoms: The 'pathology' of motivation and goal-directed behaviour. *Trends in Neurosciences*, *23*, 412–417.

Buchsbaum, M. S., Wu, J., Siegel, B. V., et al. (1997). Effect of sertraline on regional metabolic rate in patients with affective disorder. *Biological Psychiatry*, *41*, 15–22.

Bush, G., Luu, P., & Posner, M. I. (2000). Cognitive and emotional influences in anterior cingulate cortex. *Trends in Cognitive Sciences*, *4*, 215–222.

Cairns, H., Olfield, R. C., Pennybacker, J. B., & Whitteridge, D. (1941). Akinetic mutism with an epidermoid cyst of the 3rd ventricle. *Brain*, *64*, 273–290.

Cantello, R., Aguggia, M., Gilli, M., et al. (1989). Major depression in Parkinson's disease and the mood response to intravenous methylphenidate: Possible role of the "hedonic" dopamine synapse. *Journal of Neurology, Neurosurgery, and Psychiatry*, *52*, 724–731.

Carlson, J. M., Foti, D., Mujica-Parodi, L. R., Harmon-Jones, E., & Hajcak, G. (2011). Ventral striatal and medial prefrontal BOLD activation is correlated with reward-related electrocortical activity: A combined ERP and fMRI study. *NeuroImage*, *57*, 1608–1616.

Castrioto, A., Lhommee, E., Moro, E., & Krack, P. (2014). Mood and behavioural effects of subthalamic stimulation in Parkinson's disease. *Lancet Neurology*, *13*, 287–305.

Chatterjee, A., & Fahn, S. (2002). Methylphenidate treats apathy in Parkinson's disease. *The Journal of Neuropsychiatry and Clinical Neurosciences*, *14*, 461–462.

Corcoran, C., Wong, M. L., & O'Keane, V. (2004). Bupropion in the management of apathy. *Journal of Psychopharmacology*, *18*, 133–135.

Cubo, E., Benito-Leon, J., Coronell, C., & Armesto, D. (2012). Clinical correlates of apathy in patients recently diagnosed with Parkinson's disease: The ANIMO study. *Neuroepidemiology*, *38*, 48–55.

Cummings, J. L., Mega, M., Gray, K., Rosenberg-Thompson, S., Carusi, D. A., & Gornbein, J. (1994). The neuropsychiatric inventory: Comprehensive assessment of psychopathology in dementia. *Neurology*, *44*, 2308–2314.

Czernecki, V., Schupbach, M., Yaici, S., et al. (2008). Apathy following subthalamic stimulation in Parkinson disease: A dopamine responsive symptom. *Movement Disorders*, *23*, 964–969.

Devos, D., Moreau, C., Maltete, D., et al. (2014). Rivastigmine in apathetic but dementia and depression-free patients with Parkinson's disease: A double-blind, placebo-controlled, randomised clinical trial. *Journal of Neurology, Neurosurgery, and Psychiatry*, *85*, 668–674.

Drapier, D., Drapier, S., Sauleau, P., et al. (2006). Does subthalamic nucleus stimulation induce apathy in Parkinson's disease? *Journal of Neurology*, *253*, 1083–1091.

Drijgers, R. L., Aalten, P., Winogrodzka, A., Verhey, F. R., & Leentjens, A. F. (2009). Pharmacological treatment of apathy in neurodegenerative diseases: A systematic review. *Dementia and Geriatric Cognitive Disorders*, *28*, 13–22.

Drijgers, R. L., Dujardin, K., Reijnders, J. S., Defebvre, L., & Leentjens, A. F. (2010). Validation of diagnostic criteria for apathy in Parkinson's disease. *Parkinsonism & Related Disorders*, *16*, 656–660.

Drui, G., Carnicella, S., Carcenac, C., et al. (2014). Loss of dopaminergic nigrostriatal neurons accounts for the motivational and affective deficits in Parkinson's disease. *Molecular Psychiatry*, *19*, 358–367.

Dujardin, K., Sockeel, P., Delliaux, M., Destee, A., & Defebvre, L. (2009). Apathy may herald cognitive decline and dementia in Parkinson's disease. *Movement Disorders*, *24*, 2391–2397.

Dujardin, K., Sockeel, P., Devos, D., et al. (2007). Characteristics of apathy in Parkinson's disease. *Movement Disorders*, *22*, 778–784.

Eckert, T., Tang, C., & Eidelberg, D. (2007). Assessment of the progression of Parkinson's disease: A metabolic network approach. *Lancet Neurology*, *6*, 926–932.

Etkin, A., Egner, T., & Kalisch, R. (2011). Emotional processing in anterior cingulate and medial prefrontal cortex. *Trends in Cognitive Sciences*, *15*, 85–93.

Fisher, C. M. (1983). Honored guest presentation: Abulia minor vs. agitated behavior. *Clinical Neurosurgery*, *31*, 9–31.

Funkiewiez, A., Ardouin, C., Caputo, E., et al. (2004). Long term effects of bilateral subthalamic nucleus stimulation on cognitive function, mood, and behaviour in Parkinson's disease. *Journal of Neurology, Neurosurgery, and Psychiatry*, *75*, 834–839.

Goetz, C. G., Emre, M., & Dubois, B. (2008). Parkinson's disease dementia: Definitions, guidelines, and research perspectives in diagnosis. *Annals of Neurology*, *64*(Suppl. 2), S81–S92.

Goodwin, G. M. (1997). Neuropsychological and neuroimaging evidence for the involvement of the frontal lobes in depression. *Journal of Psychopharmacology*, *11*, 115–122.

Gray, T. S. (1999). Functional and anatomical relationships among the amygdala, basal forebrain, ventral striatum, and cortex. An integrative discussion. *Annals of the New York Academy of Sciences*, *877*, 439–444.

Groenewegen, H. J., Wright, C. I., & Uylings, H. B. (1997). The anatomical relationships of the prefrontal cortex with limbic structures and the basal ganglia. *Journal of Psychopharmacology*, *11*, 99–106.

Gross, C. G. (2009). Three before their time: Neuroscientists whose ideas were ignored by their contemporaries. *Experimental Brain Research*, *192*, 321–334.

Habib, M. (2004). Athymhormia and disorders of motivation in basal ganglia disease. *The Journal of Neuropsychiatry and Clinical Neurosciences*, *16*, 509–524.

Halliday, A. M., & Nathan, P. W. (1961). Methylphenidate in parkinsonism. *British Medical Journal*, *1*(5240), 1652–1655.

Herrmann, N., Rothenburg, L. S., Black, S. E., et al. (2008). Methylphenidate for the treatment of apathy in Alzheimer disease: Prediction of response using dextroamphetamine challenge. *Journal of Clinical Psychopharmacology*, *28*, 296–301.

Huang, C., Ravdin, L. D., Nirenberg, M. J., et al. (2013). Neuroimaging markers of motor and nonmotor features of Parkinson's disease: An 18f fluorodeoxyglucose positron emission computed tomography study. *Dementia and Geriatric Cognitive Disorders*, *35*, 183–196.

Hurt, C., Bhattacharyya, S., Burns, A., et al. (2008). Patient and caregiver perspectives of quality of life in dementia. An investigation of the relationship to behavioural and psychological symptoms in dementia. *Dementia and Geriatric Cognitive Disorders*, *26*, 138–146.

Ilango, A., Kesner, A. J., Keller, K. L., Stuber, G. D., Bonci, A., & Ikemoto, S. (2014). Similar roles of substantia nigra and ventral tegmental dopamine neurons in reward and aversion. *The Journal of Neuroscience*, *34*, 817–822.

Isella, V., Melzi, P., Grimaldi, M., et al. (2002). Clinical, neuropsychological, and morphometric correlates of apathy in Parkinson's disease. *Movement Disorders*, *17*, 366–371.

Jahanshahi, M. (1998). Willed action and its impairments. *Cognitive Neuropsychology*, *15*, 483–533.

Kirsch-Darrow, L., Fernandez, H. H., Marsiske, M., Okun, M. S., & Bowers, D. (2006). Dissociating apathy and depression in Parkinson disease. *Neurology*, *67*, 33–38.

Kirsch-Darrow, L., Marsiske, M., Okun, M. S., Bauer, R., & Bowers, D. (2011). Apathy and depression: Separate factors in Parkinson's disease. *Journal of the International Neuropsychological Society*, *17*, 1058–1066.

Krack, P., Pollak, P., Limousin, P., et al. (1998). Subthalamic nucleus or internal pallidal stimulation in young onset Parkinson's disease. *Brain*, *121*(Pt. 3), 451–457.

Kuhl, J., & Atkinson, J. W. (1986). *Motivation, thought, and action*. New York: Praeger Publishers.

Kulisevsky, J., Avila, A., Barbanoj, M., Antonijoan, R., Berthier, M. L., & Gironell, A. (1996). Acute effects of levodopa on neuropsychological performance in stable and fluctuating Parkinson's disease patients at different levodopa plasma levels. *Brain*, *119*(Pt. 6), 2121–2132.

Kulisevsky, J., Pagonabarraga, J., & Martinez-Corral, M. (2009). Changes in artistic style and behaviour in Parkinson's disease: Dopamine and creativity. *Journal of Neurology*, *256*, 816–819.

Kulisevsky, J., Pagonabarraga, J., Pascual-Sedano, B., Garcia-Sanchez, C., & Gironell, A. (2008). Prevalence and correlates of neuropsychiatric symptoms in Parkinson's disease without dementia. *Movement Disorders*, *23*, 1889–1896.

Laatu, S., Karrasch, M., Martikainen, K., & Marttila, R. (2013). Apathy is associated with activities of daily living ability in Parkinson's disease. *Dementia and Geriatric Cognitive Disorders*, *35*, 249–255.

Lanctot, K. L., Chau, S. A., Herrmann, N., et al. (2014). Effect of methylphenidate on attention in apathetic AD patients in a randomized, placebo-controlled trial. *International Psychogeriatrics*, *26*, 239–246.

Lanctot, K. L., Moosa, S., Herrmann, N., et al. (2007). A SPECT study of apathy in Alzheimer's disease. *Dementia and Geriatric Cognitive Disorders*, *24*, 65–72.

Laplane, D., Baulac, M., Widlocher, D., & Dubois, B. (1984). Pure psychic akinesia with bilateral lesions of basal ganglia. *Journal of Neurology, Neurosurgery, and Psychiatry*, *47*, 377–385.

Laplane, D., & Dubois, B. (2001). Auto-activation deficit: A basal ganglia related syndrome. *Movement Disorders*, *16*, 810–814.

Le Jeune, F., Drapier, D., Bourguignon, A., et al. (2009). Subthalamic nucleus stimulation in Parkinson disease induces apathy: A PET study. *Neurology*, *73*, 1746–1751.

Leentjens, A. F., Dujardin, K., Marsh, L., et al. (2008a). Apathy and anhedonia rating scales in Parkinson's disease: Critique and recommendations. *Movement Disorders*, *23*, 2004–2014.

Leentjens, A. F., Dujardin, K., Marsh, L., et al. (2008b). Anxiety rating scales in Parkinson's disease: Critique and recommendations. *Movement Disorders*, *23*, 2015–2025.

Leentjens, A. F., Koester, J., Fruh, B., Shephard, D. T., Barone, P., & Houben, J. J. (2009). The effect of pramipexole on mood and motivational symptoms in Parkinson's disease: A meta-analysis of placebo-controlled studies. *Clinical Therapeutics*, *31*, 89–98.

Levy, R., & Dubois, B. (2006). Apathy and the functional anatomy of the prefrontal cortex-basal ganglia circuits. *Cerebral Cortex*, *16*, 916–928.

Lhommée, E., Batir, A., Quesada, J. L., Ardouin, C., Fraix, V., & Seigneuret, E. (2014). Dopamine and the biology of creativity: Lessons from Parkinson's disease. *Frontiers in Neurology*, *5*, 55.

Lhommee, E., Klinger, H., Thobois, S., et al. (2012). Subthalamic stimulation in Parkinson's disease: Restoring the balance of motivated behaviours. *Brain*, *135*, 1463–1477.

Liao, C., Feng, Z., Zhou, D., et al. (2012). Dysfunction of fronto-limbic brain circuitry in depression. *Neuroscience*, *201*, 231–238.

Luk, K. C., Kehm, V., Carroll, J., et al. (2012). Pathological alpha-synuclein transmission initiates Parkinson-like neurodegeneration in nontransgenic mice. *Science*, *338*, 949–953.

Marin, R. S. (1991). Apathy: A neuropsychiatric syndrome. *The Journal of Neuropsychiatry and Clinical Neurosciences*, *3*, 243–254.

Marin, R. S. (1996). Apathy: Concept, syndrome, neural mechanisms, and treatment. *Seminars in Clinical Neuropsychiatry*, *1*, 304–314.

Marin, R. S., Biedrzycki, R. C., & Firinciogullari, S. (1991). Reliability and validity of the apathy evaluation scale. *Psychiatry Research*, *38*, 143–162.

Marin, R. S., Fogel, B. S., Hawkins, J., Duffy, J., & Krupp, B. (1995). Apathy: A treatable syndrome. *The Journal of Neuropsychiatry and Clinical Neurosciences*, 7, 23–30.

Marshall, G. A., Monserratt, L., Harwood, D., Mandelkern, M., Cummings, J. L., & Sultzer, D. L. (2007). Positron emission tomography metabolic correlates of apathy in Alzheimer disease. *Archives of Neurology*, *64*, 1015–1020.

Martinez-Corral, M., Pagonabarraga, J., Llebaria, G., et al. (2010). Facial emotion recognition impairment in patients with Parkinson's disease and isolated apathy. *Parkinson's Disease*, *2010*, 930627.

Martinez-Horta, S., Pagonabarraga, J., Fernandez de Bobadilla, R., Garcia-Sanchez, C., & Kulisevsky, J. (2013). Apathy in Parkinson's disease: More than just executive dysfunction. *Journal of the International Neuropsychological Society*, *19*, 571–582.

Martinez-Horta, S., Riba, J., de Bobadilla, R. F., et al. (2014). Apathy in Parkinson's disease: Neurophysiological evidence of impaired incentive processing. *The Journal of Neuroscience*, *34*, 5918–5926.

Martinez-Horta, S., Sampedro, F., Pagonabarraga, J., et al. (2016). Non-demented Parkinson's disease patients with apathy show decreased grey matter volume in key executive and reward-related nodes. *Brain Imaging and Behavior*. [Epub ahead of print].

Martinez-Martin, P., Rodriguez-Blazquez, C., Kurtis, M. M., & Chaudhuri, K. R. (2011). The impact of non-motor symptoms on health-related quality of life of patients with Parkinson's disease. *Movement Disorders*, *26*, 399–406.

Mayberg, H. S., Brannan, S. K., Tekell, J. L., et al. (2000). Regional metabolic effects of fluoxetine in major depression: Serial changes and relationship to clinical response. *Biological Psychiatry*, *48*, 830–843.

Mayberg, H. S., Liotti, M., Brannan, S. K., et al. (1999). Reciprocal limbic-cortical function and negative mood: Converging PET findings in depression and normal sadness. *The American Journal of Psychiatry*, *156*, 675–682.

McCutcheon, J. E., Ebner, S. R., Loriaux, A. L., & Roitman, M. F. (2012). Encoding of aversion by dopamine and the nucleus accumbens. *Frontiers in Neuroscience*, *6*, 137.

Mega, M. S., & Cohenour, R. C. (1997). Akinetic mutism: Disconnection of frontal-subcortical circuits. *Neuropsychiatry, Neuropsychology, and Behavioral Neurology*, *10*, 254–259.

Mega, M. S., Cummings, J. L., Salloway, S., & Malloy, P. (1997). The limbic system: An anatomic, phylogenetic, and clinical perspective. *The Journal of Neuropsychiatry and Clinical Neurosciences*, *9*, 315–330.

Mellers, B. A., Schwartz, A., & Cooke, A. D. (1998). Judgment and decision making. *Annual Review of Psychology*, *49*, 447–477.

Merrilees, J., Dowling, G. A., Hubbard, E., Mastick, J., Ketelle, R., & Miller, B. L. (2013). Characterization of apathy in persons with frontotemporal dementia and the impact on family caregivers. *Alzheimer Disease and Associated Disorders*, *27*, 62–67.

Migneco, O., Benoit, M., Koulibaly, P. M., et al. (2001). Perfusion brain SPECT and statistical parametric mapping analysis indicate that apathy is a cingulate syndrome: A study in Alzheimer's disease and nondemented patients. *NeuroImage*, *13*, 896–902.

Mogenson, G. J., Jones, D. L., & Yim, C. Y. (1980). From motivation to action: Functional interface between the limbic system and the motor system. *Progress in Neurobiology*, *14*, 69–97.

Monchi, O., Petrides, M., Doyon, J., Postuma, R. B., Worsley, K., & Dagher, A. (2004). Neural bases of set-shifting deficits in Parkinson's disease. *The Journal of Neuroscience*, *24*, 702–710.

Monchi, O., Petrides, M., Mejia-Constain, B., & Strafella, A. P. (2007). Cortical activity in Parkinson's disease during executive processing depends on striatal involvement. *Brain*, *130*, 233–244.

Nagaratnam, N., Nagaratnam, K., Ng, K., & Diu, P. (2004). Akinetic mutism following stroke. *Journal of Clinical Neuroscience*, *11*, 25–30.

Naville, F. (1922). Etudes sur les complications et les sequelles mentales de l'encephalite epidemique. *Encepale*, *17*, 369–375.

Padala, P. R., Burke, W. J., Shostrom, V. K., et al. (2010). Methylphenidate for apathy and functional status in dementia of the Alzheimer type. *The American Journal of Geriatric Psychiatry*, *18*, 371–374.

Parent, A., & Hazrati, L. N. (1995). Functional anatomy of the basal ganglia. I. The cortico-basal ganglia-thalamo-cortical loop. *Brain Research. Brain Research Reviews*, *20*, 91–127.

Pluck, G. C., & Brown, R. G. (2002). Apathy in Parkinson's disease. *Journal of Neurology, Neurosurgery, and Psychiatry*, *73*, 636–642.

Pondal, M., Marras, C., Miyasaki, J., et al. (2013). Clinical features of dopamine agonist withdrawal syndrome in a movement disorders clinic. *Journal of Neurology, Neurosurgery, and Psychiatry*, *84*, 130–135.

Ray Chaudhuri, K., Martinez-Martin, P., Antonini, A., et al. (2013). Rotigotine and specific non-motor symptoms of Parkinson's disease: Post hoc analysis of RECOVER. *Parkinsonism & Related Disorders*, *19*, 660–665.

Ray, N. J., Miyasaki, J. M., Zurowski, M., et al. (2012). Extrastriatal dopaminergic abnormalities of DA homeostasis in Parkinson's patients with medication-induced pathological gambling: A [$^{11}$C] FLB-457 and PET study. *Neurobiology of Disease*, *48*, 519–525.

Remy, P., Doder, M., Lees, A., Turjanski, N., & Brooks, D. (2005). Depression in Parkinson's disease: Loss of dopamine and noradrenaline innervation in the limbic system. *Brain*, *128*, 1314–1322.

Robert, P. H., Clairet, S., Benoit, M., et al. (2002). The apathy inventory: Assessment of apathy and awareness in Alzheimer's disease, Parkinson's disease and mild cognitive impairment. *International Journal of Geriatric Psychiatry*, *17*, 1099–1105.

Robert, G., Le Jeune, F., Dondaine, T., et al. (2014). Apathy and impaired emotional facial recognition networks overlap in Parkinson's disease: A PET study with conjunction analyses. *Journal of Neurology, Neurosurgery, and Psychiatry*, *85*, 1153–1158.

Robert, P., Onyike, C. U., Leentjens, A. F., et al. (2009). Proposed diagnostic criteria for apathy in Alzheimer's disease and other neuropsychiatric disorders. *European Psychiatry*, *24*, 98–104.

Robinson, O. J., Cools, R., Carlisi, C. O., Sahakian, B. J., & Drevets, W. C. (2012). Ventral striatum response during reward and punishment reversal learning in unmedicated major depressive disorder. *The American Journal of Psychiatry*, *169*, 152–159.

Schiehser, D. M., Liu, L., Lessig, S. L., et al. (2013). Predictors of discrepancies in Parkinson's disease patient and caregiver ratings of apathy, disinhibition, and executive dysfunction before and after diagnosis. *Journal of the International Neuropsychological Society*, *19*, 295–304.

Schrag, A., Barone, P., Brown, R. G., et al. (2007). Depression rating scales in Parkinson's disease: Critique and recommendations. *Movement Disorders*, *22*, 1077–1092.

Sockeel, P., Dujardin, K., Devos, D., Deneve, C., Destee, A., & Defebvre, L. (2006). The Lille apathy rating scale (LARS), a new instrument for detecting and quantifying apathy: Validation in Parkinson's disease. *Journal of Neurology, Neurosurgery, and Psychiatry*, *77*, 579–584.

Starkstein, S. E. (2012). Apathy in Parkinson's disease: Diagnostic and etiological dilemmas. *Movement Disorders*, *27*, 174–178.

Starkstein, S. E., Mayberg, H. S., Preziosi, T. J., Andrezejewski, P., Leiguarda, R., & Robinson, R. G. (1992). Reliability, validity, and clinical correlates of apathy in Parkinson's disease. *The Journal of Neuropsychiatry and Clinical Neurosciences*, *4*, 134–139.

Starkstein, S. E., Merello, M., Jorge, R., Brockman, S., Bruce, D., & Power, B. (2009). The syndromal validity and nosological position of apathy in Parkinson's disease. *Movement Disorders*, *24*, 1211–1216.

Steeves, T. D., Miyasaki, J., Zurowski, M., et al. (2009). Increased striatal dopamine release in Parkinsonian patients with pathological gambling: A [$^{11}$C] raclopride PET study. *Brain*, *132*, 1376–1385.

Stone, E. A., Lin, Y., & Quartermain, D. (2008). A final common pathway for depression? Progress toward a general conceptual framework. *Neuroscience and Biobehavioral Reviews*, *32*, 508–524.

Thobois, S., Ardouin, C., Lhommee, E., et al. (2010). Non-motor dopamine withdrawal syndrome after surgery for Parkinson's disease: Predictors and underlying mesolimbic denervation. *Brain*, *133*, 1111–1127.

Thobois, S., Lhommee, E., Klinger, H., et al. (2013). Parkinsonian apathy responds to dopaminergic stimulation of D2/D3 receptors with piribedil. *Brain*, *136*, 1568–1577.

van Eimeren, T., Pellecchia, G., Cilia, R., et al. (2010). Drug-induced deactivation of inhibitory networks predicts pathological gambling in PD. *Neurology*, *75*, 1711–1716.

van Reekum, R., Stuss, D. T., & Ostrander, L. (2005). Apathy: Why care? *The Journal of Neuropsychiatry and Clinical Neurosciences*, *17*, 7–19.

Vijayaraghavan, L., Krishnamoorthy, E. S., Brown, R. G., & Trimble, M. R. (2002). Abulia: A delphi survey of British neurologists and psychiatrists. *Movement Disorders*, *17*, 1052–1057.

Wongpakaran, N., van Reekum, R., Wongpakaran, T., & Clarke, D. (2007). Selective serotonin reuptake inhibitor use associates with apathy among depressed elderly: A case-control study. *Annals of General Psychiatry*, *6*, 7.

Ye, Z., Hammer, A., Camara, E., & Munte, T. F. (2011). Pramipexole modulates the neural network of reward anticipation. *Human Brain Mapping*, *32*, 800–811.

Zahodne, L. B., Bernal-Pacheco, O., Bowers, D., et al. (2012). Are selective serotonin reuptake inhibitors associated with greater apathy in Parkinson's disease? *The Journal of Neuropsychiatry and Clinical Neurosciences*, *24*, 326–330.

Zahodne, L. B., Gongvatana, A., Cohen, R. A., Ott, B. R., & Tremont, G. (2013). Are apathy and depression independently associated with longitudinal trajectories of cortical atrophy in mild cognitive impairment? *The American Journal of Geriatric Psychiatry*, *21*, 1098–1106.

Zahodne, L. B., Marsiske, M., Okun, M. S., & Bowers, D. (2012). Components of depression in Parkinson disease. *Journal of Geriatric Psychiatry and Neurology*, *25*, 131–137.

Zahodne, L. B., & Tremont, G. (2013). Unique effects of apathy and depression signs on cognition and function in amnestic mild cognitive impairment. *International Journal of Geriatric Psychiatry*, *28*, 50–56.

Zahodne, L. B., Young, S., Kirsch-Darrow, L., et al. (2009). Examination of the Lille apathy rating scale in Parkinson disease. *Movement Disorders*, *24*, 677–683.